



Comparative efficacy of pomegranate juice, pomegranate rind powder extract and BHT as antioxidants in cooked chicken patties

B.M. Naveena ^{*}, A.R. Sen, S. Vaithiyanathan, Y. Babji, N. Kondaiah

National Research Centre on Meat, Chengicherla, P.B. No. 19, Uppal Post, Hyderabad, Andhra Pradesh 500039, India

ARTICLE INFO

Article history:

Received 16 April 2008

Received in revised form 10 June 2008

Accepted 11 June 2008

Keywords:

Pomegranate juice

Rind powder

BHT

Natural phenolics

Chicken patties

ABSTRACT

A study was carried out to evaluate the antioxidant potential of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT) in cooked chicken patties during refrigerated storage. Freshly minced chicken meats were assigned to one of the following four treatments: control (meat treated with no antioxidants); 10 mg equivalent PJ phenolics per 100 g meat; 10 mg equivalent RP phenolics per 100 g meat; 10 mg BHT per 100 g meat. The patties formed from the minced meats were grilled for 20 min and stored under aerobically at 4 °C for 15 days. Total phenolic content (as tannic acid equivalent) significantly ($P < 0.05$) increased from 152 in control to 195 and 224 µg/g in PJ and RP patties. Addition of PJ or RP did not affect any of the sensory attributes. The TBARS values were significantly ($P < 0.05$) reduced from 1.272 in control patties to 0.896, 0.763 and 0.203 mg malonaldehyde per kg samples in BHT, PJ and RP patties, respectively. The RP treatment substantially inhibited ($P < 0.01$) lipid oxidation in cooked chicken patties to a much greater extent than BHT treatment. The PJ or RP at a level of 10 mg equivalent phenolics/100 g meat would be sufficient to protect chicken patties against oxidative rancidity for periods longer than the most commonly used synthetic antioxidant like BHT.

© 2008 Elsevier Ltd. All rights reserved.

1. Introduction

The meat industry is increasingly seeking natural solutions to minimize oxidative rancidity and extend the shelf-life of meat products rather than artificial additives, such as butylated hydroxyl anisole (BHA), butylated hydroxyl toluene (BHT), propyl gallate and tertiary butyl hydroxyl queneone (TBHQ). However, the synthetic antioxidants currently used have been found to exhibit various health effects. The continuous use of such synthetic chemicals may cause health hazards such as teratogenic and carcinogenic effects in laboratory animals and primates (Hathway, 1966). There has been a growing interest in natural ingredients because they have greater application for increasing consumer acceptability, palatability, stability and shelf-life of food products. Consequently, search for natural additives, especially of plant origin, has notably increased in recent years. Waste products from processing of fruit and vegetables offer a practical and economic source of potent antioxidants that could replace synthetic preservatives. Compounds obtained from natural sources such as grains, oilseeds, spices, honey, fruits and vegetables have been investigated (Chen,

Chan, Ma, Fung, & Wang, 1996; Naveena, Sen, Vaithiyanathan, Muthukumar, & Babji, 2007).

Pomegranate is an important source of bioactive compounds and has been used in folk medicine for many centuries. Most pomegranate fruit parts are known to possess enormous antioxidant activity. In India, pomegranate arils are used as such or are made into juice. Pomegranate juice has been demonstrated to be high in antioxidant activity and is effective in prevention of atherosclerosis, low-density lipoprotein oxidation, prostate cancer, platelet aggregation and various cardiovascular diseases (Adhami & Mukhtar, 2006). Ozkal and Dinc (1994) reported the presence of tannins, anthocyanins and flavonoids in pomegranate rind. Pomegranate peel is a rich source of tannins and other phenolic compounds (Ozkal & Dinc, 1994).

While substantial data exists in favor of use of polyphenols from green tea, rosemary, berry fruits, thyme, sage, and other herbs as natural antioxidants (Bozkurt, 2006; Pokorny, 1991), interest in the antioxidant properties of polyphenols from pomegranate has recently emerged (Naveena, Sen, Kingsly, Singh, & Kondaiah, in press). The utilization of pomegranate fruits for meat processing and its potential health benefits are not well understood. In this study pomegranate juice and pomegranate rind powder was extracted using water. Total phenolic content of extracts was determined and the antioxidant properties of these extracts containing different concentrations of phenolic compounds were assessed in cooked chicken patties.

^{*} Corresponding author. Tel.: +91 40 25502951; fax: +91 40 27201672.

E-mail address: naveenlpt@rediffmail.com (B.M. Naveena).

2. Materials and methods

2.1. Materials

Fresh meat from manually deboned chicken carcasses was obtained from local poultry processing plants in Hyderabad. Meat samples were stored at 4 °C for approximately 4 h before use. Fresh meat samples were obtained separately for each of the five replications. Butylated hydroxyl toluene (BHT) was obtained from MERK, Mumbai, India.

2.2. Preparation of pomegranate juice (PJ)

Fresh pomegranate fruits (Kabul variety) obtained from a local super market were washed and cut into four pieces. The seeds/arils were separated manually and ground in a mixer for 30 s and filtered through muslin cloth. The freshly prepared juice was used for each replication after analyzing for pH, total phenolic content, DPPH radical scavenging activity and reducing power.

2.3. Preparation of pomegranate rind powder and extract (RP)

Mature and healthy pomegranate fruits were washed and cut manually to separate the seeds and rind. Rind was cut into small pieces using a sharp knife and dried in an air circulatory tray drier (Narang Scientific Works, New Delhi, India) at 60 °C for 48 h. Dried pieces were cooled and powdered in a heavy duty grinder and sieved using a 60 mesh sieve and packed and stored at room temperature in high density poly ethylene bags until extraction. About 20 g of dried rind powder was mixed with 500 ml boiled distilled water and left for 5 min. The extract was obtained by filtration and analyzed for pH, total phenolic content, reducing power and DPPH radical scavenging activity. Freshly prepared extract was used for each replication.

2.4. Preparation of chicken patties

Chicken samples (about 6 kg meat from both leg and breast for each replication) were minced twice (13 mm plate followed by 6 mm plates using a meat mincer (SCHARFEN, Model X70, 58413 Witten, West Germany). After mincing, the samples were assigned to one of the following four treatments (Table 1): Control (meat without any antioxidant); PJ 10 (10 mg equivalent PJ phenolics per 100 g meat); RP 10 (10 mg equivalent RP phenolics per 100 g meat); BHT (100 mg BHT per 100 g meat). Sodium chloride (1% w/w) dissolved in distilled water was added to all samples. The BHT was dissolved in 5 ml sunflower oil before addition and the same quantity of oil was added to other samples to maintain uniformity. The volume of PJ and RP extract added were replaced with distilled water in control and BHT samples. Immediately after adding all ingredients samples were thoroughly hand-mixed.

Table 1

Ingredients for chicken patties formulated with pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT)

Treatments	Meat (g)	PJ (ml)	RP (ml)	BHT (ppm)	Salt (g)	Oil (ml)	Distilled water (ml)
Control patties	100	–	–	–	1	5	8
PJ patties	100	8	–	–	1	5	–
RP patties	100	–	1	–	1	5	7
BHT patties	100	–	–	100	1	5	8

PJ: 8 ml contains 10 mg equivalent PJ phenolics.

RP: 1 ml contains 10 mg equivalent RP phenolics.

After mixing, minced chicken (100 g portions) was formed into patties and cooked in a microwave oven (Model MC-767 w/w, LG Electronics India Pvt. Ltd., New Delhi, India) under grilling for around 20 min until the internal temperature reached 80 °C. After cooling to room temperature the patties were aerobically packaged in a low density polyethylene pouches and stored at 4 °C for 15 days and analyzed for total phenolic content, pH, instrumental color, sensory attributes and thiobarbituric acid reactive substances (TBARS). All the parameters and sensory evaluation were done only on day 0 except the TBARS values, which were analyzed throughout the storage.

2.5. Analysis of samples

2.5.1. DPPH radical scavenging activity

The ability to scavenge 1,1-diphenyl 1-2-picrylhydrazyl (DPPH) radical by PJ, RP and BHT was estimated by the method of Singh, Murthy, and Jayaprakasha (2002). The PJ, RP and BHT (50 and 100 µg) diluted with 0.1 M Tris–HCl buffer (pH 7.4) was mixed with 1 ml of DPPH (250 µM) with vigorous shaking. The reaction mixture was stored in the dark at room temperature for 20 min and the absorbance was measured at 517 nm (Model: UV–VIS 5704 SS, ECIL, Hyderabad, India). The scavenging activity was calculated by the following equation:

$$\text{Scavenging activity \%} = \frac{(\text{Absorbance}_{\text{Blank}} - \text{Absorbance}_{\text{Sample}})}{\text{Absorbance}_{\text{Blank}}} \times 100.$$

2.5.2. Measurement of reducing power

The reducing power was quantified by the method described by Jayaprakasha, Sing, and Sakariah (2001). The 50–100 µg phenolics from PJ, RP and BHT were mixed with 2.5 ml phosphate buffer (200 µM, pH 6.6) and incubated with 2.5 ml potassium ferricyanide (1% w/v) at 50 °C for 20 min. At the end of incubation, 2.5 ml of 10% trichloroacetic acid solution was added and centrifuged at 9700g for 10 min. The supernatant was mixed with 5 ml distilled water and 1 ml ferric chloride (0.1% w/v) solution. The absorbance was measured at 700 nm (Model: UV–VIS 5704 SS, ECIL, Hyderabad, India). Increase in absorbance of the reaction indicated the reducing power of the sample.

2.5.3. Total phenolics

The PJ, RP and cooked chicken patties were analyzed for total phenolics using the Folin–Ciocalteus (F–C) assay (Escarpa & Gonzalez, 2001) with slight modifications. Five gram of cooked patty was homogenized with 25 ml of 70% acetone and kept overnight for extraction at refrigeration temperature. Suitable aliquots of extracts were taken in a test tube and the volume was made to 0.5 ml with distilled water followed by the addition of 0.25 ml F–C (1 N) reagent and 1.25 ml sodium carbonate solution (20%). The tubes were vortexed and the absorbance recorded at 725 nm (Model: UV–VIS 5704 SS, ECIL, Hyderabad, India) after 40 min. The amount of total phenolics was calculated as tannic acid equivalent from the calibration curve using standard tannic acid solution (0.1 mg/ml).

2.5.4. pH and cooking yield

The pH of the cooked patty was determined by blending 10 g sample with 50 ml distilled water for 60 s in a homogenizer (Model: MICRA D8-Si, ART Moderne Labortechnik, D-79379 Mullheim, Germany). The pH values were measured using a standardized electrode attached to a digital pH meter (Thermo Orion, Model 420A+, USA). Cooking yield was determined by dividing cooked product weight by the raw uncooked weight and multiplying by 100.

2.5.5. Instrumental color

Instrumental color analysis was performed using a Hunter lab Miniscan XE Plus colorimeter (Hunter Associates Laboratory Inc., Reston, VA, USA) at day 0 with 25 mm aperture set for illumination D65, 10° standard observer angle. CIE L^* (lightness), a^* (redness) and b^* (yellowness) were measured on the outer and internal surface of cooked chicken patties from five randomly chosen spots. Hue angle ($\tan^{-1} b^*/a^*$) and chroma ($(a^{*2} + b^{*2})^{1/2}$) were calculated according to Hunter and Harold (1987).

2.5.6. Thiobarbituric acid reactive substances

The thiobarbituric acid reactive substances (TBARS) value (mg malonaldehyde/kg) of the chicken patties was determined using the extraction method described by Witte, Krauze, and Bailey (1970) with slight modifications as the slurry was centrifuged at 3000g for 10 min (Centurion Scientific, Model K2 series, UK) instead of filtration through Whatman No. 42.

2.5.7. Sensory evaluation

An experienced panel of 8–10 members evaluated the cooked chicken patties on day 0 of storage. The panelists rated each sample for three characteristics (appearance, juiciness and overall palatability) on an 8-point descriptive scale (Keeton, 1983) and three characteristics (off-odour, sweet flavour and chicken flavour) on a 5-point rating scale (Leheska et al., 2006; Vasavada, Dwivedi, & Cornforth, 2006). Patties were warmed in a microwave oven for

20 s just before sensory evaluation and coded samples were served at room temperature in separate booths. Water was served for cleansing the mouth between samples.

2.5.8. Statistical analysis

All data were analyzed using SPSS (SPSS version 13.0 for windows; SPSS, Chicago, IL, USA). The cooking yield, pH, instrumental color and sensory attributes were analyzed using one-way ANOVA. A 4×4 factorial design with five replicates was employed for storage data (TBARS), with treatments and storage time as main effects using two-way ANOVA. The least significant difference (LSD) was calculated at $P < 0.05$.

3. Results and discussion

The scavenging effects of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT) at different concentrations of total phenolic content on the DPPH radical is shown in Fig. 1. The DPPH was used as a free radical to evaluate antioxidant activity present in natural sources (Schwarz, Bertelson, & Nissen, 2001). Increasing the concentration of RP and BHT increased ($P < 0.05$) the radical scavenging activity compared to PJ. Negi and Jayaprakasha (2003) have reported a sharp increase in radical scavenging activity with an increase in the concentration of pomegranate peel extracts up to 25 ppm after which little increase in radical scavenging activity was observed.

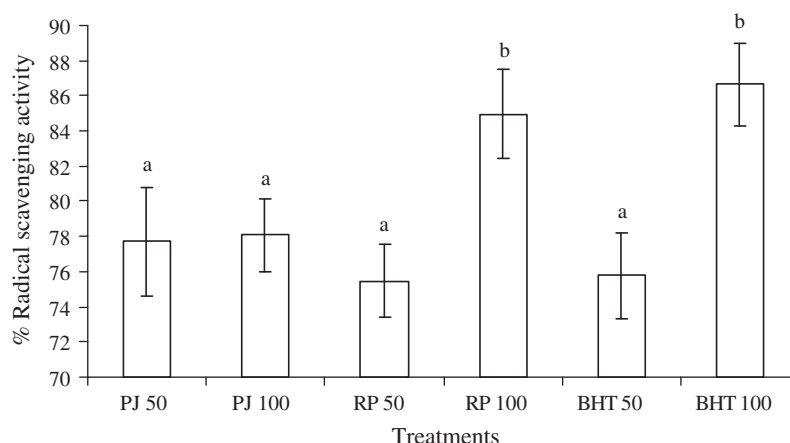


Fig. 1. The DPPH (1,1-diphenyl-2-picrylhydrazyl) radical scavenging activity of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT) at different concentrations. Least-square means and standard error bars are indicated ($n = 5$). Means with no common letters differ significantly ($P < 0.05$). PJ50, 50 μ g PJ phenolics; PJ100, 100 μ g PJ phenolics; RP50, 50 μ g RP phenolics; RP100, 100 μ g RP phenolics; BHT50, 50 μ g BHT; BHT100, 100 μ g BHT.

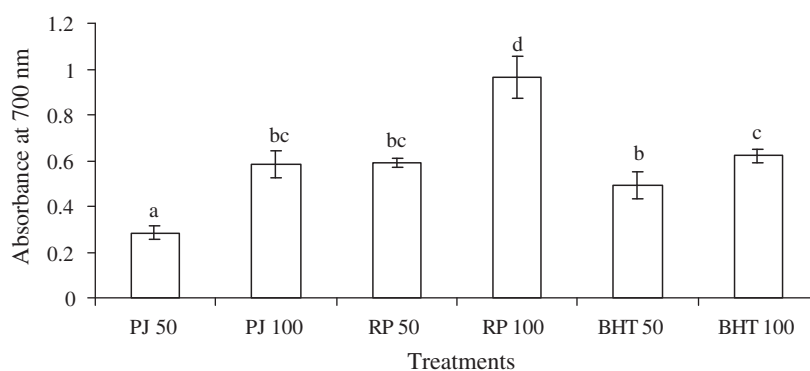


Fig. 2. The reducing power of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT) at different concentrations. Least-square means and standard error bars are indicated ($n = 5$). Means with no common letters differ significantly ($P < 0.05$). PJ50, 50 μ g PJ phenolics; PJ100, 100 μ g PJ phenolics; RP50, 50 μ g RP phenolics; RP100, 100 μ g RP phenolics; BHT50, 50 μ g BHT; BHT100, 100 μ g BHT.

Fig. 2 shows the reducing powers of different concentration of phenolic compounds in PJ, RP and BHT using the potassium ferricyanide method. There was a significant ($P < 0.05$) increase in reducing power of PJ, RP and BHT with increase in concentration. The RP 100 had highest ($P < 0.01$) reducing power compared to the others. Negi and Jayaprakash (2003) have also reported a significant increase in reducing power of pomegranate peel extracts with increase in concentration from 50 to 400 ppm. Reducing properties are generally associated with the presence of reductones (Duh, 1998). Gordon (1990) reported that the antioxidative action of reductones is based on the breaking of free radical chains by the donation of hydrogen atom.

The pH of raw and cooked chicken patties, cooking yield and total phenolic content of cooked patty is given in Table 2. The raw meat pH significantly ($P < 0.05$) reduced in BHT patties compared to the others. The total phenolic content of cooked chicken patties was significantly ($P < 0.05$) higher in PJ and RP patties compared to control and BHT patties. The higher level of phenolics may indicate this product is nutritionally enhanced due to the fruit juice and rind powder that was added (Leheska et al., 2006).

Preliminary work with evaluation of instrumental color in raw patties with added PJ and RP revealed no significant difference (data not shown). All Treatments reduced ($P < 0.05$) the Hunterlab L^* (lightness) values on the surface of cooked patties compared to control patties (Table 3). Significant increase in Hunter lab a^* (redness) values was observed in RP and BHT patties compared to others. However, no difference was observed in the internal a^* values between samples. This might be because cooking at high temperature (80 °C) might have completely denatured the myoglobin and hence no difference was observed in internal a^* values. With the addition of RP the chicken patties became slightly darker, which might have resulted in lower L^* values. Addition of PJ changed the chicken meat patties from pale raw to grayish color with lower instrumental values. Mitsumoto, O'grady, Kerry, and Buckley (2005) have reported the discoloration of chicken meat patties with addition of natural antioxidants like tea catechins. It should be noted that some other factors could have an effect on meat color parameters such as the fineness of mincing and consequently surface reflection properties. In contrast to instrumental color values sensory evaluation scores (Table 4) did not reveal any significant

Table 2

The pH (raw and cooked), cooking yield and total phenolic content of cooked chicken patties as affected by addition of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT)

Treatment	Raw pH	Cooked pH	Cooking yield (%)	Total phenolics (as tannic acid eq) $\mu\text{g/g}$
Control patties	6.03 $\pm$ 0.03 ^b	6.11 $\pm$ 0.14	85.56 $\pm$ 8.16	152.00 $\pm$ 19.55 ^a
PJ patties	6.00 $\pm$ 0.02 ^{ab}	6.03 $\pm$ 0.19	85.88 $\pm$ 7.00	195.00 $\pm$ 10.60 ^b
RP patties	6.01 $\pm$ 0.03 ^{ab}	6.10 $\pm$ 0.13	84.07 $\pm$ 6.68	224.00 $\pm$ 27.01 ^c
BHT patties	5.98 $\pm$ 0.01 ^a	6.12 $\pm$ 0.15	85.39 $\pm$ 7.68	161.00 $\pm$ 18.16 ^a

Number of observations = 5.

Mean values in the same column bearing the same superscript do not differ significantly ($P < 0.05$).

Table 3

Instrumental color values of cooked chicken patties as affected by addition of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT)

Treatment	L^*	a^*	b^*	Hue	Chroma
<i>Surface color</i>					
Control patties	72.60 $\pm$ 4.81 ^c	3.31 $\pm$ 1.56 ^a	21.32 $\pm$ 5.11 ^b	80.97 $\pm$ 5.65 ^c	21.61 $\pm$ 2.28 ^b
PJ patties	62.14 $\pm$ 4.97 ^a	3.36 $\pm$ 1.02 ^a	16.31 $\pm$ 4.91 ^a	71.40 $\pm$ 6.71 ^b	16.66 $\pm$ 3.81 ^a
RP patties	63.12 $\pm$ 3.78 ^a	4.01 $\pm$ 1.50 ^b	16.89 $\pm$ 5.13 ^a	76.30 $\pm$ 5.88 ^a	17.38 $\pm$ 3.22 ^a
BHT patties	70.17 $\pm$ 5.12 ^b	3.97 $\pm$ 1.55 ^b	22.17 $\pm$ 5.34 ^b	79.69 $\pm$ 6.17 ^{bc}	22.54 $\pm$ 3.11 ^b
<i>Internal color</i>					
Control patties	74.15 $\pm$ 2.80 ^b	2.21 $\pm$ 0.95	17.81 $\pm$ 2.92 ^b	82.80 $\pm$ 6.29	17.94 $\pm$ 1.95 ^b
PJ patties	66.53 $\pm$ 3.11 ^a	2.34 $\pm$ 0.58	14.53 $\pm$ 3.21 ^a	80.56 $\pm$ 5.86	14.71 $\pm$ 2.56 ^a
RP patties	68.61 $\pm$ 3.98 ^a	2.42 $\pm$ 0.84	15.85 $\pm$ 2.55 ^{ab}	81.36 $\pm$ 4.98	16.03 $\pm$ 3.78 ^{ab}
BHT patties	74.10 $\pm$ 3.85 ^b	2.38 $\pm$ 0.51	18.04 $\pm$ 3.10 ^b	82.87 $\pm$ 5.66	18.19 $\pm$ 2.54 ^b

Number of observations = 25.

Mean values in the same column bearing the same superscript do not differ significantly ($P < 0.05$).

L^* , lightness; a^* , redness; b^* , yellowness; Hue, $\tan^{-1} b^*/a^*$; Chroma, $(a^{*2} + b^{*2})^{1/2}$.

Table 4

Sensory evaluation scores of cooked chicken patties as affected by addition of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT)

Treatment	Appearance ^a	Juiciness ^b	Off-odor ^c	Sweet flavour ^c	Chicken flavour ^c	Overall palatability ^d
Control patties	7.26 $\pm$ 0.11	7.16 $\pm$ 0.07	5.00 $\pm$ 0.00	5.00 $\pm$ 0.00	1.03 $\pm$ 0.03	7.66 $\pm$ 0.10
PJ patties	7.10 $\pm$ 0.18	7.40 $\pm$ 0.13	5.00 $\pm$ 0.00	4.99 $\pm$ 0.04	1.06 $\pm$ 0.04	7.50 $\pm$ 0.10
RP patties	7.03 $\pm$ 0.15	7.43 $\pm$ 0.13	5.00 $\pm$ 0.00	5.00 $\pm$ 0.02	1.06 $\pm$ 0.04	7.53 $\pm$ 0.10
BHT patties	7.23 $\pm$ 0.04	7.30 $\pm$ 0.09	5.00 $\pm$ 0.00	5.00 $\pm$ 0.06	1.05 $\pm$ 0.04	7.60 $\pm$ 0.02

Number of observations = 25.

Mean values in the same column bearing the same superscript do not differ significantly ($P < 0.05$).

^a Appearance: 1 = extremely poor to 8 = excellent.

^b Juiciness: 1 = extremely dry to 8 = extremely juicy.

^c Off-odor, sweet flavour, chicken flavour: 1 = extremely intense odor or flavour to 5 = no flavour or odor.

^d Overall palatability: 1 = extremely palatable to 8 = extremely unpalatable.

Table 5

Thiobarbituric acid reactive substances (TBARS) values of cooked chicken patties as affected by addition of pomegranate juice (PJ), rind powder extract (RP) and butylated hydroxyl toluene (BHT) during refrigerated storage (mg of malonaldehyde kg⁻¹ meat)

Treatment	0th day	5th day	10th day	15th day
Control	0.307 ± 0.08 ^{cA}	0.635 ± 0.09 ^{cB}	1.016 ± 0.30 ^{cC}	1.272 ± 0.13 ^{cD}
PJ	0.271 ± 0.08 ^{bA}	0.266 ± 0.03 ^{bA}	0.485 ± 0.16 ^{bB}	0.763 ± 0.16 ^{bC}
RP	0.103 ± 0.03 ^{aA}	0.098 ± 0.01 ^{aA}	0.140 ± 0.03 ^{aA}	0.203 ± 0.04 ^{aB}
BHT	0.187 ± 0.05 ^{abA}	0.388 ± 0.17 ^{abB}	0.498 ± 0.31 ^{abB}	0.896 ± 0.12 ^{abC}

^{a–d}Treatments within the same storage conditions (column-wise) with different superscripts are significantly different ($P < 0.05$). ^{A–D}Storage conditions within the same treatment (row-wise) with different superscript are significantly different ($P < 0.05$); each value is a mean ± SE of five replicates.

difference in appearance between control and treated samples. No significant difference was observed in off-odour, sweet flavour, chicken flavour and overall palatability scores at any level. All the products were equally acceptable as evidenced by the overall acceptability scores.

Effect of PJ, RP and BHT treatment on thiobarbituric acid reactive substances (TBARS) values in cooked chicken patties are shown in Table 5. All the treatments significantly ($P < 0.05$) reduced the TBARS values throughout storage compared to control. The lipid oxidation inhibition effect was highest ($P < 0.05$) in RP compared to PJ and BHT at all storage times. However, no difference was observed between PJ and BHT treated patties. The TBARS values significantly ($P < 0.05$) increased in control patties throughout storage, however in treated patties the values increased ($P < 0.05$) up to the 5th day of storage only. The increase in TBARS values in RP treated samples was very slow and remained lowest (<0.2 mg malonaldehyde/kg sample) up to 15 days. The large amount of phenolics contained in rind powder extract may cause its strong antioxidant ability (Li et al., 2006). The pomegranate peel phenolics may act in a similar fashion as reductones by donating electrons and reacting with free radicals to convert them to more stable products and terminate free radical chain reactions (Negi & Jayaprakasha, 2003). The data indicate that the marked antioxidant activity of RP seems to be the result of their radical scavenging activity and reducing power. Natural, antioxidants are believed to break free radical chains of oxidation by donation of a hydrogen from the phenolic groups, thereby forming a stable end product (Sherwin, 1978).

4. Conclusion

Pomegranate rind powder (RP) and pomegranate juice (PJ) have substantial amounts of phenolic compounds. The RP phenolics significantly inhibited lipid oxidation in cooked chicken patties to a much greater extent than PJ and BHT. Addition of RP at a level of 10 mg equivalent phenolics/100 g meat would be sufficient to protect chicken patties against oxidative rancidity for periods longer than BHT. The meat industry could use these fruits or fruit byproducts as a potential source of phenolics as they have immense nutraceutical value and can be used to produce functional meat

products of commercial interest. Further research is required to isolate and identify their active components.

References

- Adhami, V. M., & Mukhtar, H. (2006). Polyphenols from green tea and pomegranate for prevention of prostate cancer. *Free Radical Research*, 40, 1095–1104.
- Bozkurt, H. (2006). Utilization of natural antioxidants: Green tea extract and *Thymra spicata* oil in Turkish dry-fermented sausage. *Meat Science*, 73, 2–450.
- Chen, Z. Y., Chan, P. T., Ma, H. M., Fung, K. P., & Wang, J. (1996). Antioxidative effect of ethanol green tea extracts on oxidation of canola oil. *Journal of the American Oil Chemists' Society*, 73, 375–380.
- Duh, P.-D. (1998). Antioxidant activity of Budrock (*Arctium lappa* L.): Its scavenging effects on free radical and active oxygen. *Journal of the American Oil Chemists Society*, 75, 455–461.
- Escarpa, A., & Gonzalez, M. C. (2001). Approach to the content of total extractable phenolic compounds from different food samples by comparison of chromatographic and spectrophotometric methods. *Anal Chim Acta*, 427, 119–127.
- Gordon, M. F. (1990). The mechanism of antioxidant action in vitro. In B. J. F. Hudson (Ed.), *Food antioxidants* (pp. 1–18). London: Elsevier Applied Science.
- Hathway, D. E. (1966). Metabolic fate in animals of hindered phenolic antioxidants in relation to their safety evaluation and antioxidants function. *Advances in Food Research*, 15, 1.
- Hunter, R. S., & Harold, R. W. (1987). *The measurement of appearance*. New York: John Wiley and Sons.
- Jayaprakasha, G. K., Sing, R. P., & Sakariah, K. K. (2001). Antioxidant activity of grape seed (*Vitis vinifera*) extracts on peroxidation models in-vitro. *Food Chemistry*, 73, 285–290.
- Keeton, J. T. (1983). Effect of fat and NaCl/ phosphate levels on the chemical and sensory properties of pork patties. *Journal of Food Science*, 41, 878–881.
- Leheska, J. M., Boyce, J., Brooks, J. C., Hoover, L. C., Thompson, L. D., & Miller, F. (2006). Sensory attributes and phenolic content of precooked pork breakfast sausage with fruit purees. *Journal of Food Science*, 71, S249–S252.
- Li, Y., Guo, C., Yang, J., Wei, J., Xu, J., & Cheng, S. (2006). Evaluation of antioxidant properties of pomegranate peel extract in comparison with pomegranate pulp extract. *Food Chemistry*, 96, 254–260.
- Mitsumoto, M., O'grady, M. N., Kerry, J. P., & Buckley, D. J. (2005). Addition of tea catechins and vitamin C on sensory evaluation, colour and lipid stability during chilled storage in cooked or raw beef and chicken patties. *Meat Science*, 69, 773–779.
- Naveena, B. M., Sen, A. R., Kingsly, R. P., Singh, D. B., & Kondaiah, N. (in press). Antioxidant activity of pomegranate rind powder extract in cooked chicken patties. *International Journal of Food Science and Technology*.
- Naveena, B. M., Sen, A. R., Vaithyanathan, S., Muthukumar, M., & Babji, Y. (2007). Microwave cooking properties of chicken patties containing honey and vitamin-C. *Journal of Food Science and Technology*, 44, 505–508.
- Negi, P. S., & Jayaprakasha, G. K. (2003). Antioxidant and antibacterial activities of *Punica granatum* peel extracts. *Journal of Food Science*, 68, 1473–1477.
- Ozkal, N., & Dinc, S. (1994). Evaluation of the pomegranate (*Punica granatum* L.) peels from the standpoint of pharmacy. *Ankara Univ Eczacilik Fak Derg*, 22, 21–29.
- Pokorny, J. (1991). Natural antioxidants for food use. *Trends in Food Science and Technology*, 2, 223–227.
- Schwarz, K., Bertelson, G., & Nissen, L. R. (2001). Investigation of plant extracts for the protection of processed foods against lipid oxidation. Comparison of antioxidant assays based on radical scavenging, lipid oxidation and analysis of the principal antioxidant compounds. *European Food Research and Technology*, 212, 319–328.
- Sherwin, E. R. (1978). Oxidation and antioxidants in fats and oil processing. *Journal of the American Oil Chemists Society*, 55, 809–815.
- Singh, R. P., Murthy, K. N. C., & Jayaprakasha, G. K. (2002). Studies on the antioxidant activity of pomegranate (*Punica granatum*) peel and seed extracts using in-vitro models. *Journal of Agricultural and Food Chemistry*, 50, 81–86.
- Vasavada, M. N., Dwivedi, S., & Cornforth, D. (2006). Evaluation of garam masala spices and phosphates as antioxidants in cooked ground beef. *Journal of Food Science*, 71, C292–C297.
- Witte, V. C., Krauze, G. F., & Bailey, M. E. (1970). A new extraction method for determining 2-thiobarbituric acid values of pork and beef during storage. *Journal of Food Science*, 35, 582–585.